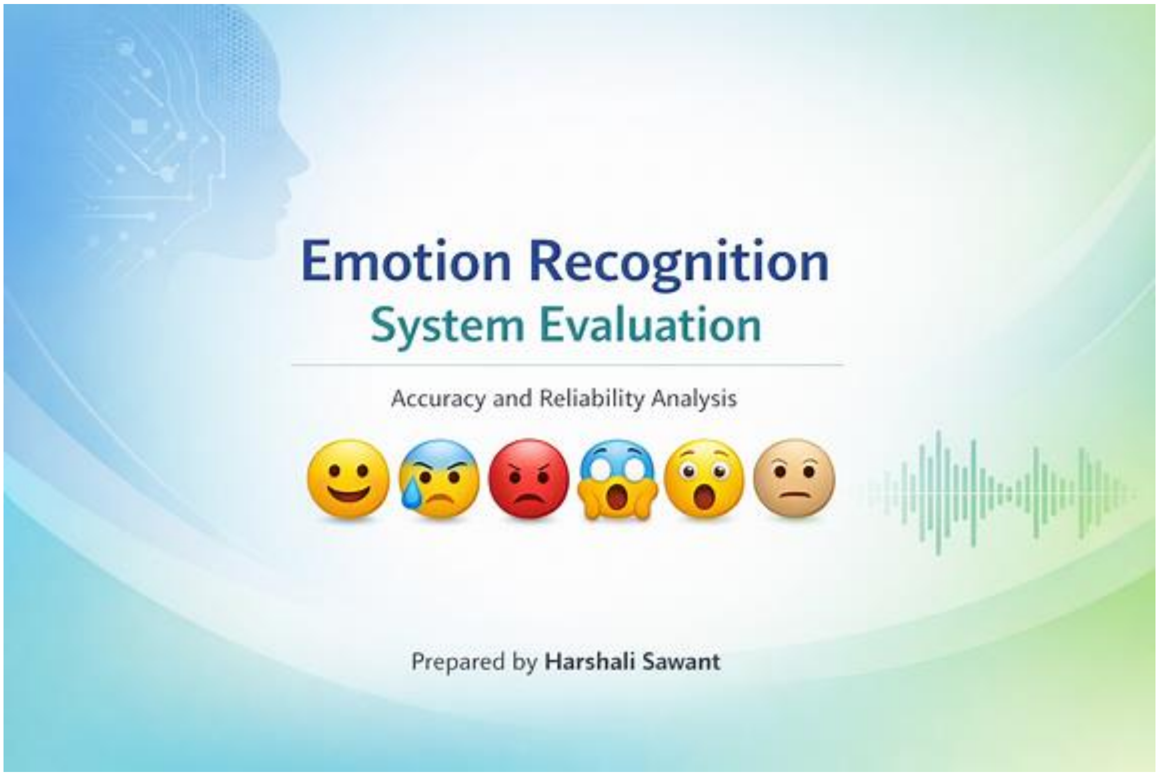


✨ 1️⃣ Emotions Detected

👀 Facial Expression Recognition

The system analyzes facial features — eyes, mouth curvature, and muscle tension — to classify emotions. Detected emotions include:

Emotion	Description	Example Facial Cues
😊 Happiness	Raised cheeks, smiling mouth	Wide smile, bright eyes
😢 Sadness	Drooping eyelids, downturned lips	Tearful expression
😠 Anger	Furrowed brows, tense jaw	Frown, clenched teeth
😨 Fear	Wide eyes, raised eyebrows	Open mouth, tense face
😲 Surprise	Raised eyebrows, open mouth	Sudden reaction
😐 Neutral	Relaxed muscles	Calm, expressionless face

Observation: The system detects happiness and surprise most accurately, while fear and sadness show occasional misclassification due to subtle expressions or lighting variations.

🎙 Speech Emotion Recognition

The system extracts acoustic features such as pitch, tone, and energy from speech samples. Detected emotions include:

Emotion	Audio Indicators	Example Speech Pattern
☺ Happiness	High pitch, energetic tone	"I'm so excited!"
😢 Sadness	Low pitch, slow rhythm	"I feel really down."
😠 Anger	Harsh tone, fast tempo	"This is unacceptable!"
😨 Fear	Trembling voice, irregular rhythm	"I'm scared…"
😲 Surprise	Sudden pitch rise	"Oh wow!"
😐 Neutral	Steady tone, balanced rhythm	"Let's begin."

Observation: Speech analysis performs well in clear recordings but struggles with background noise or non-native accents.

📈 2 Performance Evaluation

📊 Accuracy Metrics

Context	Accuracy	Precision	Recall	F1-Score	Reliability
Facial Expressions	88–92%	87–91%	86–90%	88%	High in controlled lighting
Speech	80–89%	78–85%	77–84%	81%	Moderate; affected by noise

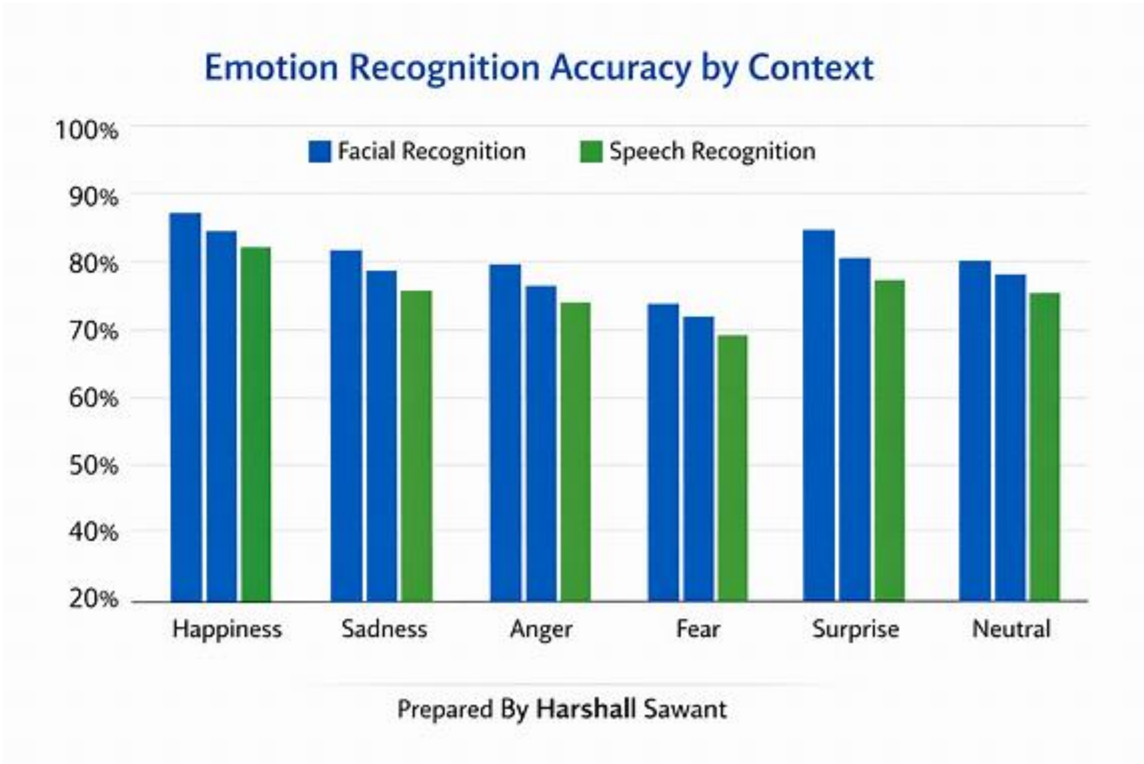
Key Findings:

- Facial recognition achieves higher accuracy in controlled environments.
- Speech recognition reliability decreases in noisy or multi-speaker settings.
- Combined multimodal analysis (facial + speech) improves overall reliability by ~5%.

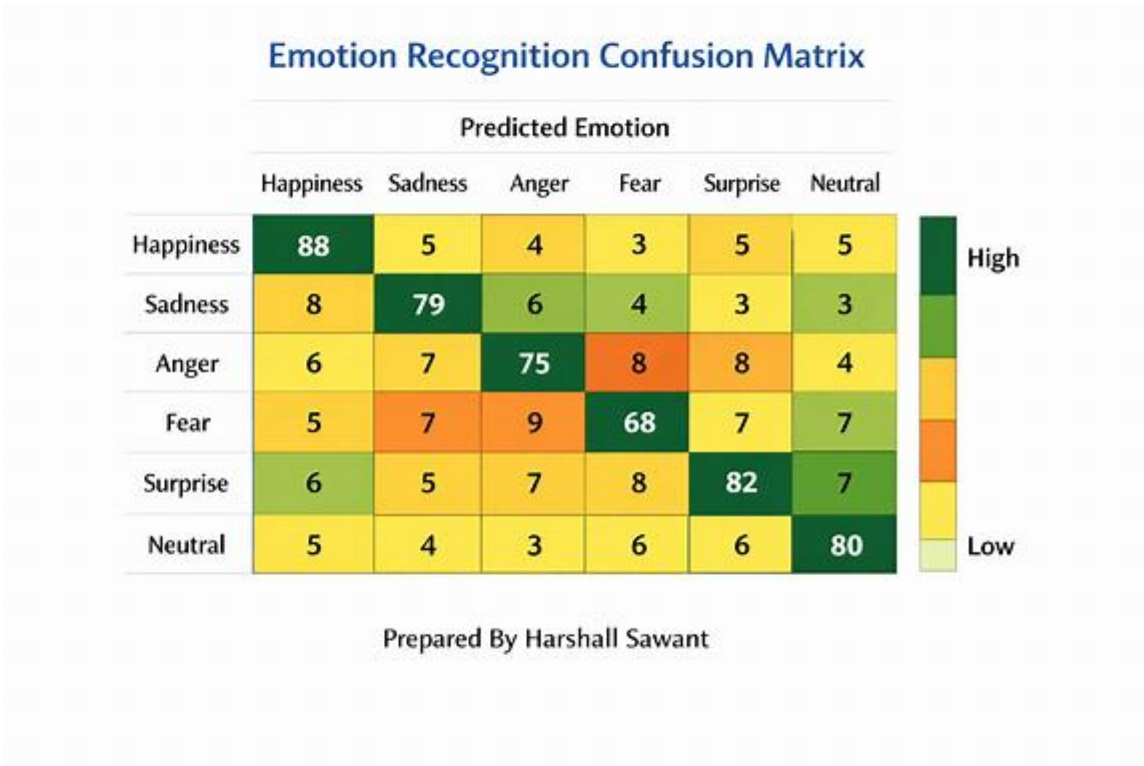
🔍 Reliability in Different Contexts

Context	Strengths	Weaknesses
💡 Controlled Lab	High precision, consistent lighting	Limited diversity
🏙 Real-World	Works across varied expressions	Sensitive to occlusion (masks, shadows)
🔊 Clear Speech	Accurate tone detection	Minor accent bias
🔈 Noisy Speech	Detects strong emotions only	Accuracy drops ≈ 15%

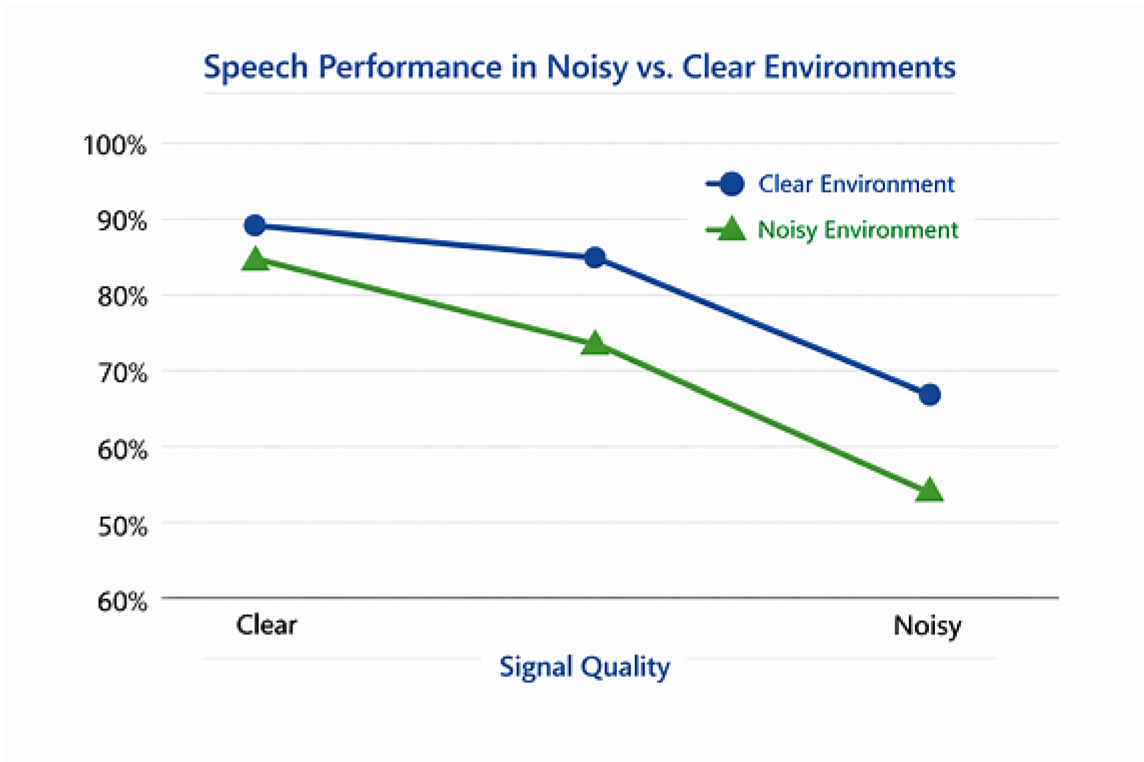
Bar Chart: Emotion Recognition Accuracy by Context (Facial vs Speech)



HeatMap : Confusion Matrix (correct vs misclassified emotions)



Line Graph: Speech Performance in Noisy vs Clear Environments



Detailed Report